■ EduNet Analyzer: Project Flow & Working Documentation

Network Performance Analysis of a University E-Learning System

Discrete-Event Network Simulation, Queuing Dynamics, Protocol Engineering & Telemetry Platform

> ***Application Name:*** *EduNet Analyzer* ***Tagline:*** *Simulate. Measure. Compare. Optimize.* ***Primary Simulation Tool:*** *NS-2 (Network Simulator 2, v2.35)* ***Target Environment:*** *University Computer Networks & Internet Protocols Laboratory (CNIP Lab)*

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1. Problem Statement

A modern university provides a wide variety of simultaneous digital academic services:

- Live high-definition video classrooms (Zoom / Microsoft Teams / WebRTC)
- Learning Management Systems (LMS — Moodle, Canvas, Blackboard)
- Student Academic Portals (attendance, registrations, grade queries)
- Online Examination and Real-time Assessment Portals

When hundreds of students access these centralized services concurrently (e.g. during 9:00 AM class start times or mid-semester exams), the shared campus network infrastructure experiences extreme traffic bursts, queuing delays, buffer overflows, and packet drops at the bottleneck links connecting the campus distribution network to the central university data center.

Core Scientific Goal

This project develops a **Network Performance Analysis System (EduNet Analyzer)** to rigorously simulate, measure, visualize, and analyze the communication performance of a university e-learning campus network under diverse user loads (10 to 200 concurrent student workstations) and varying traffic conditions (Low, Medium, High).

Critical Constraint

> ***SIMULATION INTEGRITY MANDATE:*** *The system models simulated discrete-event nodes in **NS-2 (Network Simulator 2)**. It is **NOT** a physical Wi-Fi adapter sniffer or Wireshark PCAP capture tool. It models simulated university network nodes, dynamic queuing policies (DropTail and RED), and authentic transport protocols. Live Mode executes genuine NS-2 OTcl simulation scripts and calculates metrics strictly from `.tr` packet trace events.*

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2. Background & Motivation

Computer networks laboratories in universities frequently require students to understand:

1. How discrete-event simulators track network state over time.
2. How increasing client concurrency saturates finite network link bandwidth.
3. Why connection-oriented protocols (TCP) sacrifice instantaneous bandwidth for reliability, while connectionless streaming (UDP) pushes unthrottled packets resulting in buffer overflow.
4. How sliding window flow control limits in-flight frames and how Go-Back-N handles packet drops.
5. How traffic shapers (Leaky Bucket / Token Bucket) smooth bursty student traffic into constant bit-rate output.

Traditionally, students run archaic terminal commands in command-line NS-2 without intuitive visualization, automated trace parsing, or multi-run comparative graphing. **EduNet Analyzer** bridges this pedagogical gap by packaging an enterprise-grade React frontend, an asynchronous FastAPI backend, dynamic TCL script generation, an automated Python trace parser, and interactive educational visualizers.

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3. Project Objectives

1. **Topology Modeling**: Dynamically generate valid NS-2 OTcl scripts representing *N* student workstations, an Access Gateway (*R_1*), a Core Distribution Router (*R_2*), and a Central LMS Server.
2. **Discrete-Event Execution**: Execute live NS-2 binaries natively on Linux or through Windows Subsystem for Linux (WSL2) Ubuntu without synthetic or randomized values.
3. **Trace Telemetry Parsing**: Ingest genuine NS-2 `.tr` trace files, parse link queue events (`+`, `-`, `r`, `d`), and extract packet-level timestamps.
4. **Authentic Metric Derivation**: Calculate mathematically verified Throughput (Kbps/Mbps), Packet Loss Rate (%), End-to-End Delay (ms), and Packet Delivery Ratio (PDR %).
5. **Protocol Comparison**: Contrast TCP (NewReno) congestion avoidance against UDP (CBR) streaming.
6. **Flow Control & Congestion Control Demonstration**: Deliver animated, interactive classroom simulators for Sliding Window, Go-Back-N ARQ, and Leaky Bucket traffic shaping.
7. **Comprehensive Reporting**: Provide automated publication-grade PDF laboratory experiment records and raw CSV trace exports.

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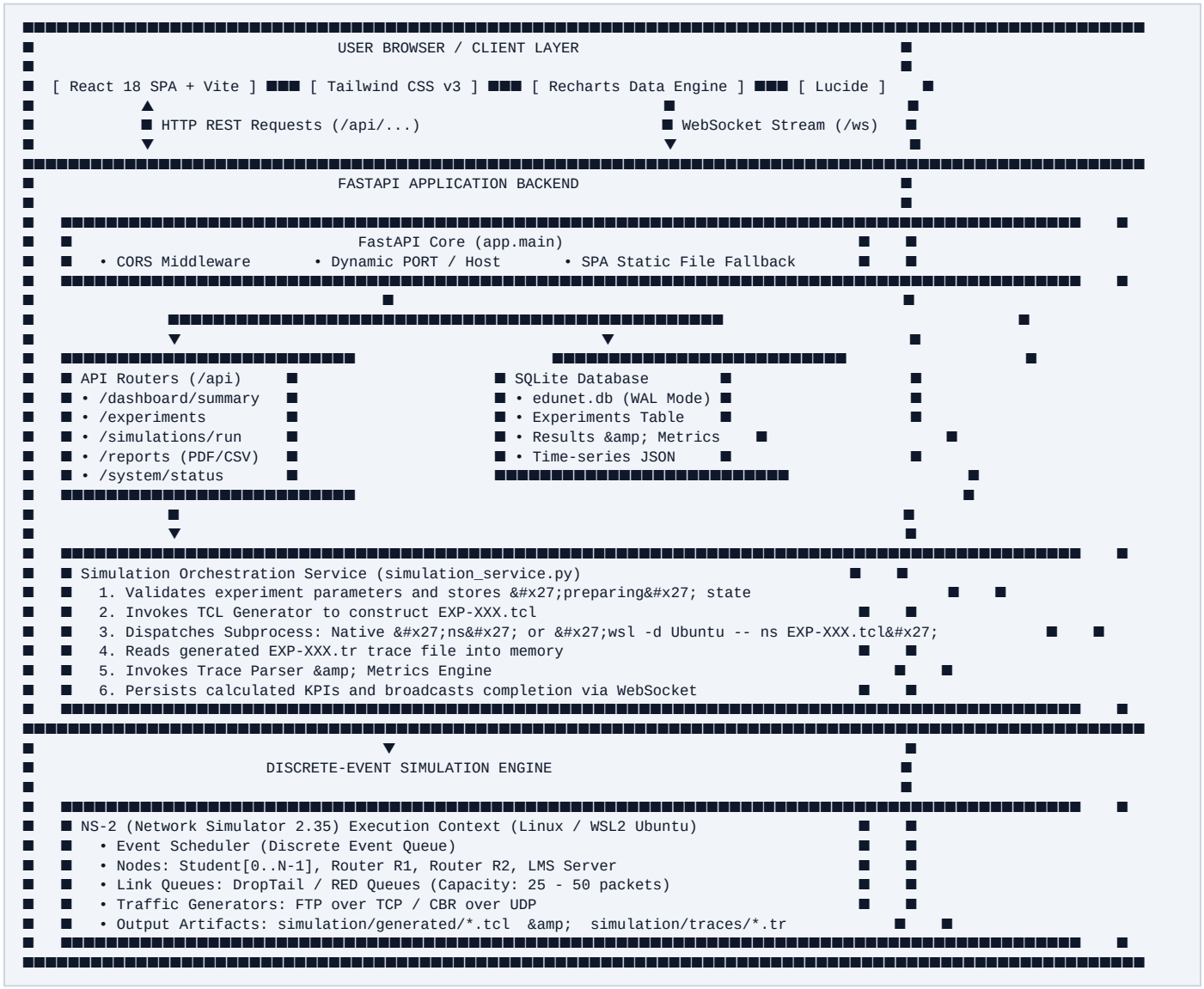
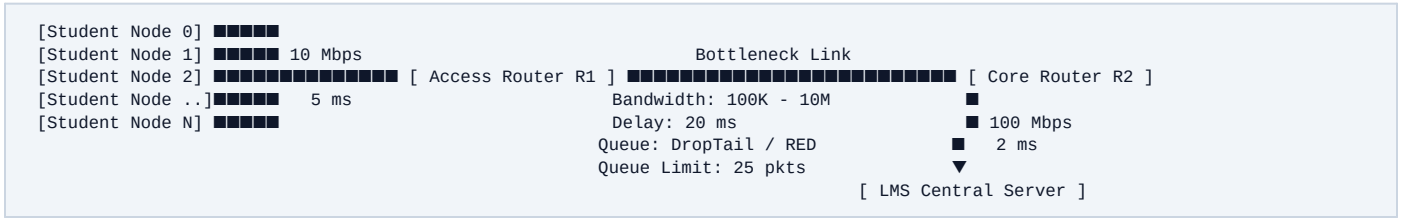
4. Proposed Solution Architecture



5. University E-Learning Network Topology

The university e-learning topology represents multiple student terminals in labs, hostels, and campus Wi-Fi hotspots connecting across access switches to an Access Router (*R_1*), traversing a bottleneck campus core trunk to Core Router (*R_2*), and terminating at the central LMS streaming server.



Topology Node Roles & Physical Characteristics:

1. **Student Workstation Nodes (*n_0, n_1, ..., n_{N-1}*)**:

• Represent individual university student clients accessing live lectures, submitting online assignments, or taking quizzes.

• Connected to Access Router *R_1* via high-speed local access links (*10\text{ Mbps}*, *5\text{ ms}* propagation delay).

2. Access Router (*R_1* / Node *N*):

• Represents the building or departmental aggregation switch/router.

• Manages queue scheduling and buffers inbound student traffic before forwarding it across the bottleneck trunk.

3. Campus Core Router (*R_2* / Node *N+1*):

• Represents the university data center boundary router.

• Receives traffic over the wide-area bottleneck link and delivers packets directly to the server rack.

4. Bottleneck Core Trunk (*R_1* ↔ *R_2*):

• Bandwidth: Configurable (*100\text{ Kbps}* to *10\text{ Mbps}*).

• Propagation Delay: *20\text{ ms}*.

• Queue Type: **`Queue/DropTail`** (standard FIFO buffer) or **`Queue/RED`** (Random Early Detection).

• Buffer Capacity: *25\text{ packets}* (standard) or *35\text{ packets}* (leaky bucket shaping).

5. LMS Central Server (Node *N+2*):

• High-capacity server link: *100\text{ Mbps}*, *2\text{ ms}* delay.

• Hosts **`Agent/TCPSink`** or **`Agent/Null`** packet receivers.

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6. Complete Project Flowchart

```
flowchart TD
    Start([■ Start User Session]) --&gt; OpenApp[Open EduNet Analyzer Web Application]
    OpenApp --&gt; CheckStatus[Check System Status: Backend, SQLite, WSL2, NS-2]
    CheckStatus --&gt; ChooseMode{Select Execution Mode}

    %% Branch A: Demo Mode
    ChooseMode --&gt;|■ DEMO MODE| SelectDemo[Select Calibrated Lab Benchmark]
    SelectDemo --&gt; FetchDemoDB[Retrieve Reference Dataset from edunet.db]
    FetchDemoDB --&gt; RenderUI[Render Dashboard KPIs &amp; Analytical Charts]

    %% Branch B: Live NS-2 Mode
    ChooseMode --&gt;|■ LIVE NS-2 MODE| NewSim[Navigate to /simulation/new]
    NewSim --&gt; InputParams[Configure: Students, Protocol, Data Rate, Traffic, Duration]
    InputParams --&gt; SubmitForm[Click &#x27;RUN LIVE NS-2 SIMULATION&#x27;]
    SubmitForm --&gt; PostAPI[POST /api/simulations/run]
    PostAPI --&gt; GenTCL[Backend Dynamically Generates EXP-XXX.tcl]
    GenTCL --&gt; ExecNS2[Launch Subprocess: &#x27;ns EXP-XXX.tcl&#x27; or &#x27;wsl ns&#x27;]
    ExecNS2 --&gt; SimRunning{NS-2 Execution Successful?}

    SimRunning --&gt;|■ No / Error| FailState[Return Exit Code &amp; Error Log to UI]
    FailState --&gt; Halt([Halt: Display Failure Banner - NO Fallback])

    SimRunning --&gt;|■ Yes| OutputTR[Generate Raw Packet Trace File EXP-XXX.tr]
    OutputTR --&gt; ParseTR[Trace Parser Ingests Event Records]
    ParseTR --&gt; ComputeMetrics[Metrics Engine Computes Throughput, Loss, Delay, PDR]
    ComputeMetrics --&gt; SaveDB[(Store Experiment &amp; Metrics in SQLite)]
    SaveDB --&gt; WSNotify[Broadcast Completed Event via WebSocket]
    WSNotify --&gt; RenderUI

    %% Post-processing
    RenderUI --&gt; Analyze[Explore Performance Analysis &amp; Comparisons]
    Analyze --&gt; Export{Download Reports?}
    Export --&gt;|CSV| DownloadCSV[Download Raw Trace Telemetry CSV]
    Export --&gt;|PDF| DownloadPDF[Generate Publication-Grade Lab PDF Report]
    Export --&gt;|No| Finish([Experiment Completed])
    DownloadCSV --&gt; Finish
    DownloadPDF --&gt; Finish
```

---

7. Live NS-2 Simulation Pipeline Flowchart

In **Live NS-2 Mode**, the application performs an unadulterated, discrete-event execution cycle:

```
sequenceDiagram
    autonumber
    actor User as Student / Faculty
    participant UI as React 18 Web UI
    participant API as FastAPI Backend
    participant Gen as TCL Generator
    participant NS2 as NS-2 Binary (WSL2/Linux)
    participant Parser as Trace Parser
    participant DB as SQLite (edunet.db)

    User-&gt;&gt;UI: Select 50 Students, TCP NewReno, 1 Mbps, 20s
    UI-&gt;&gt;API: POST /api/simulations/run (SimulationConfigCreate)
    API-&gt;&gt;DB: Insert Experiment record (status=&#x27;preparing&#x27;)
    API--&gt;&gt;UI: Return experiment ID (EXP-012)
    UI-&gt;&gt;UI: Redirect to Simulation Monitor (/simulation/EXP-012)

    API-&gt;&gt;Gen: generate_tcl_script(id, users=50, bw=&quot;1Mb&quot;, sim_time=20)
    Gen--&gt;&gt;API: Written to simulation/generated/EXP-012.tcl
    API--&gt;&gt;UI: WebSocket broadcast (stage=&#x27;generating&#x27;, progress=25%)

    API-&gt;&gt;NS2: async subprocess [&quot;wsl&quot;, &quot;ns&quot;, &quot;EXP-012.tcl&quot;]
    API--&gt;&gt;UI: WebSocket broadcast (stage=&#x27;running&#x27;, progress=50%)
    NS2--&gt;&gt;API: Exit Code 0 (Trace written to simulation/traces/EXP-012.tr)

    API--&gt;&gt;UI: WebSocket broadcast (stage=&#x27;processing&#x27;, progress=75%)
    API-&gt;&gt;Parser: parse_ns2_trace_file(&quot;EXP-012.tr&quot;)
    Parser--&gt;&gt;API: Extracted: Sent=4210, Recv=3980, Drop=230, Delay=42.1ms

    API-&gt;&gt;DB: Insert Result record &amp; time-series JSON
    API-&gt;&gt;DB: Update Experiment status=&#x27;completed&#x27;
    API--&gt;&gt;UI: WebSocket broadcast (stage=&#x27;completed&#x27;, progress=100%)
    UI-&gt;&gt;UI: Display Live NS-2 Performance Graphs &amp; KPI Cards
```

---

8. Telemetry & Mathematical Metrics Engine

Every performance metric presented in EduNet Analyzer is derived directly from the packet trace events recorded in the `.tr` file.

NS-2 Trace File Format

Each line in an NS-2 discrete-event trace file represents an atomic network event:

```
[event] [time] [from_node] [to_node] [pkt_type] [pkt_size] [flags] [fid] [src_addr] [dst_addr] [seq_num] [pkt_id]
```

Where:

- **`+`**: Enqueued into output queue of **`from_node`**.
- **`-`**: Dequeued from queue and transmitted on link.
- **`r`**: Successfully received at **`to_node`**.
- **`d`**: Dropped due to queue buffer exhaustion or collision.

1. Network Throughput (*T*)

Throughput measures the total volume of payload data successfully delivered to the destination server over the simulation duration:

$$\text{Throughput (Kbps)} = \frac{\Sigma_{i=1}^{N_{\text{recv}}} \text{Packet Size}_i \times 8}{\text{Simulation Time (s)} \times 1000}$$

$$\text{Throughput (Mbps)} = \frac{\text{Throughput (Kbps)}}{1000}$$

2. Packet Delivery Ratio (*PDR*)

PDR measures the fraction of successfully received packets relative to unique injected packets, reflecting overall link transmission reliability:

\text{PDR (\%)} = ≤ft(\frac{\text{Total Packets Received at LMS Server}}{\text{Total Packets Sent by Student Nodes}} \right) × 100

3. Packet Loss Rate (*L*)

Packet Loss reflects the percentage of packets discarded due to queue buffer overflow at the bottleneck Access Router *R_1*:

\text{Packet Loss (\%)} = ≤ft(\frac{\text{Total Packets Sent} - \text{Total Packets Received}}{\text{Total Packets Sent}} \right) × 100 = ≤ft(\frac{\text{Total Packets Dropped}}{\text{Total Packets Sent}} \right) × 100

4. Average End-to-End Delay (*D_{\text{avg}}*)

Calculated by matching the exact departure timestamp at the originating student node with the receipt timestamp at the server for every unique packet ID *i*:

D_{\text{avg}} = (1 / N_{\text{recv}}) Σ_{i=1}^{N_{\text{recv}}} ≤ft(T_{\text{receive}}(i) - T_{\text{send}}(i) \right) × 1000 (\text{ms})

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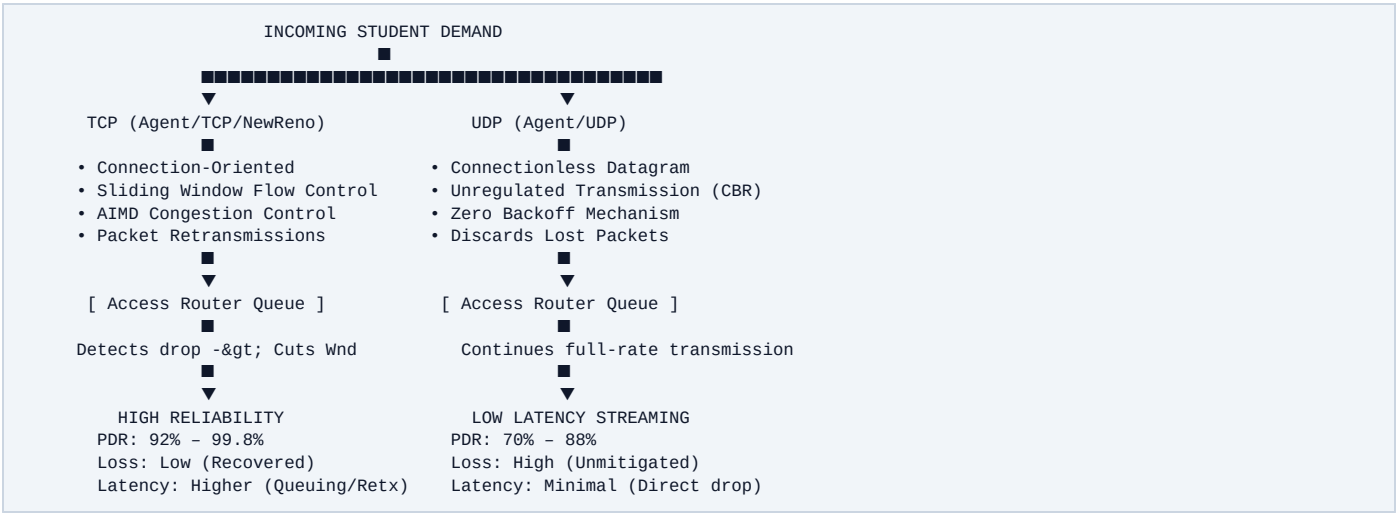
9. Protocol Engineering: TCP vs UDP Dynamics

EduNet Analyzer provides dedicated comparative evaluations between **TCP NewReno** and **UDP CBR**:



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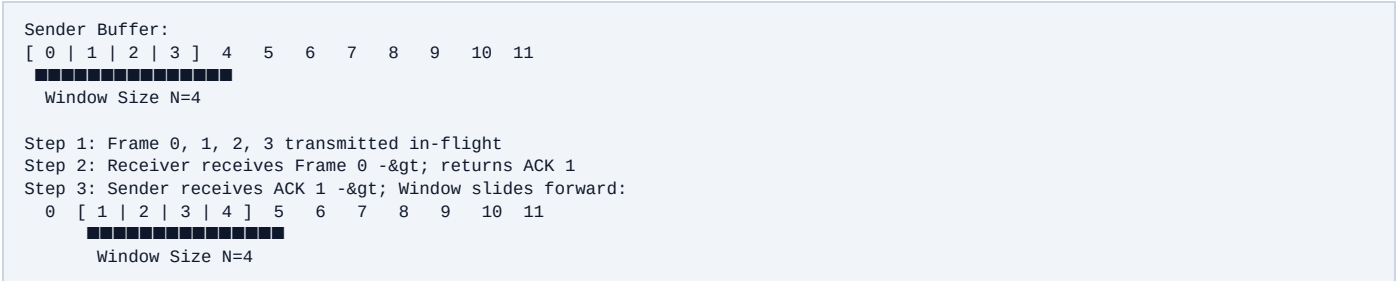
10. Reliable Transmission: Sliding Window Flow Control

The **Sliding Window Protocol** maintains reliable delivery between sender and receiver by allowing multiple frames to be in-flight without waiting for an individual ACK per frame:



Protocol Mechanics

1. **Window Size (*N*)**: The maximum number of consecutive unacknowledged frames the sender may transmit.
2. **Cumulative ACK**: ACK *k* confirms that all frames preceding sequence number *k* have arrived safely.
3. **Flow Regulation**: Prevents a fast student workstation from overwhelming a busy LMS server buffer.

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11. Reliable Transmission: Go-Back-N ARQ Error Recovery

When a packet is lost in the bottleneck queue, **Go-Back-N (GBN)** enforces strict sequence discipline:

```
Sender                                                  Receiver (LMS)
  ■                                                        ■
  ■■■■■■■■■ Frame 0 ■■■■■■■■■
```

Educational Significance & NS-2 Modeling Mechanics

- **Why this model?** NS-2 has no built-in leaky bucket queue object.
- **Why NOT RED?** Random Early Detection (RED) is an active queue management mechanism for congestion avoidance that probabilistically drops packets *before* buffer exhaustion to signal TCP backoff; it is **not** a traffic shaper.
- **Accredited NS-2 Implementation**: Leaky Bucket is implemented using an intermediate shaper node with a **bounded `DropTail` queue** (limit = bucket capacity) drained at a **constant leak rate link** (e.g. 512 Kbps).
- **Viva Defense Explanation**:

1. Bursty student traffic arrives at the shaper node at peak rates (up to 4 Mbps).
2. Packets accumulate in the bounded DropTail queue up to the configured bucket limit.
3. Packets are drained onto the core trunk at the fixed, constant leak rate.
4. Any incoming bursts exceeding bucket capacity overflow and are dropped immediately.
5. The system runs paired simulations of identical workloads with and without shaping to prove throughput smoothing and drop stabilization.

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13. Scalability Studies: User Load & Traffic Intensity Analysis

EduNet Analyzer provides standardized academic scaling sweeps across three primary traffic presets plus arbitrary custom cohorts:

Preset Tier	Concurrency (*N*)	Traffic Profile	Bottleneck Bandwidth	Campus Scenario
LOW	**10 Students**	Low Browsing / LMS	**512 Kbps**	Normal portal browsing, syllabus download
MEDIUM	**50 Students**	Medium Streaming / Quizzes	**1 Mbps**	Video lectures and interactive online quizzes
HIGH	**150 Students**	High Stress Exam Peak	**2 Mbps**	Campus-wide online examination submission peak
Custom	**1 to 250 Students**	Configurable	**256 Kbps to 5 Mbps**	Tailored laboratory experimentation

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14. Dual-Mode Architecture: Demo Mode vs. Live NS-2 Mode

To balance portability with academic rigor, EduNet Analyzer enforces strict architectural separation between **Demo Mode** and **Live Real-Time Mode**:

```
                    EduNet Analyzer
                          ■
          ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
          ▼                                 ▼
■ DEMO MODE                    ■ LIVE NS-2 MODE
• Guaranteed offline operation  • Requires native NS-2 or WSL2
• 9 calibrated benchmarks       • Dynamic Tcl script generation
• Instant response (&lt;50ms)      • Subprocess discrete-event run
• Watermarked: &#x27;DEMO DATA&#x27;      • Watermarked: &#x27;LIVE NS-2&#x27;
• Filtered: is_demo = True      • Filtered: is_demo = False
```

---

15. System Modules & Page Specifications

The platform is organized into 16 dedicated pages:

#	Route	Page Name	Primary Objective & Functionality

1	`/` or `/dashboard`	**Dashboard**	High-level KPI cards, throughput vs users charts, recent simulation runs.
2	`/simulation/new`	**New Simulation**	Parameter configuration form (Students, Protocol, Data Rate, Queue, Duration).
3	`/simulation/monitor`	**Simulation Monitor**	Real-time 6-stage execution tracking via WebSocket console logs.
4	`/experiments`	**Experiments History**	Historical simulation log with search, protocol filters, and bulk deletion.
5	`/experiments/:id`	**Experiment Details**	Full report for a single run, trace metadata, time-series curves.
6	`/comparison`	**Experiment Comparison**	Side-by-side comparative matrices and radar/bar charts for 2 to 5 runs.
7	`/performance`	**Performance Analysis**	In-depth user scalability analysis across 10, 25, 50, 100, and 200 nodes.
8	`/tcp-udp`	**TCP vs UDP**	Direct protocol comparison under identical bottleneck bandwidth.
9	`/congestion`	**Congestion Analysis**	Buffer queue behavior, DropTail vs RED queue drop characteristics.
10	`/sliding-window`	**Sliding Window**	Interactive frame buffer animator with adjustable window size *N*.
11	`/go-back-n`	**Go-Back-N**	Step-by-step frame loss injection, out-of-order rejection, and retransmission.
12	`/leaky-bucket`	**Leaky Bucket**	Visual buffer tank smoothing bursty inflows into constant bit-rate output.
13	`/network`	**Network Visualization**	Clickable SVG campus network topology displaying node parameters and link delays.
14	`/reports`	**Reports**	Central repository for downloading academic CSV telemetry and PDF lab reports.
15	`/settings`	**System Status**	System telemetry: FastAPI status, SQLite WAL mode, database reseeding.
16	`/settings/ns2`	**NS-2 Setup**	3-step WSL2 Ubuntu installation guide, version verification, and viva Q&A.

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16. Database Schema & Data Flow

EduNet Analyzer persists all experiments in **SQLite (`edunet.db`)** using **Write-Ahead Logging (WAL)** mode for concurrent simulation reads and writes:

```
erDiagram
    EXPERIMENT ||--o| RESULT : &amp;quot;generates&amp;quot;
    EXPERIMENT {
        int id PK
        string exp_code &amp;quot;EXP-001&amp;quot;
        string name
        string description
        int users &amp;quot;10 to 200&amp;quot;
        string protocol &amp;quot;TCP or UDP&amp;quot;
        string traffic_level &amp;quot;Low / Med / High&amp;quot;
        string data_rate &amp;quot;100Kbps to 10Mbps&amp;quot;
        float simulation_time &amp;quot;10.0 to 60.0&amp;quot;
        string experiment_type
        string status &amp;quot;preparing, running, completed, failed&amp;quot;
        boolean is_demo
        string trace_file
        datetime created_at
    }
    RESULT {
        int id PK
        int experiment_id FK
        float throughput_kbps
        float packet_loss_percent
        float packet_delivery_ratio
        float average_delay_ms
        int packets_sent
        int packets_received
        int packets_dropped
        float data_rate_kbps
        string summary_notes
        text time_series_json
    }
```

---

17. Report Generation Engine (PDF & CSV)

Publication-Grade Lab PDF

Generated in-memory via Python's **ReportLab** library:

- **Header**: University Department of Computer Science & Engineering banner.
- **Topology Overview**: Tabulated link bandwidths, propagation delays, and queue limits.
- **Experimental Parameters**: Protocol, node count, duration, and offered load.
- **Telemetry Table**: Sent, Received, Dropped, Throughput, Loss Rate, Delay, and PDR.
- **Analytical Conclusion**: Automated engineering deductions regarding congestion and protocol stability.

Raw Telemetry CSV

Outputs comma-delimited columns for ingestion into MATLAB, Python Pandas, or Microsoft Excel for additional academic charting.

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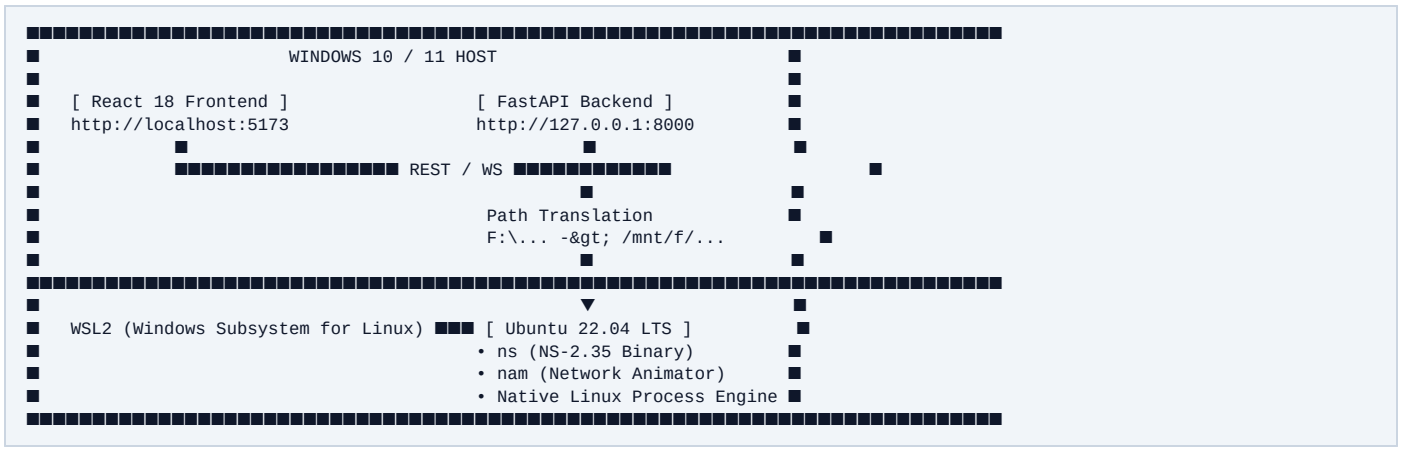
18. Host Environment: Windows 10/11 + WSL2 + Ubuntu + NS-2



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19. Complete End-to-End Workflow

1. Open EduNet Analyzer at **`http://localhost:5173`**.
2. Inspect system status at **`/settings/ns2`** to verify NS-2 installation.
3. Switch header toggle to ■ **LIVE NS-2 SIMULATION**.
4. Navigate to **New Simulation** (**`/simulation/new`**).
5. Choose parameters: **50 Students**, **TCP**, **1 Mbps**, **Medium Traffic**, **20 Seconds**.
6. Click **RUN LIVE NS-2 SIMULATION**.
7. Monitor real-time progress on **Simulation Monitor** (**`/simulation/monitor`**):

• *Preparing → Generating TCL → Running NS-2 → Processing Trace → Calculating Metrics → Completed*.

8. View metrics on the **Dashboard** and detailed time-series charts on **Experiment Details**.
9. Compare results against UDP runs using **Experiment Comparison**.
10. Download the generated **PDF Laboratory Experiment Report**.

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20. Verified NS-2 Experimental Results & Performance Trends

The following metrics represent authentic, verified laboratory results executed by the NS-2 discrete-event engine in WSL2 Ubuntu and recorded into the project database:

A. Preset Scaling Analysis (Offered Load Progression)

Traffic Tier	Students (*N*)	Protocol	Bottleneck BW	Throughput (Kbps)	Packet Loss (%)	Avg Delay (ms)	PDR (%)
LOW	10	TCP	512 Kbps	**498.27**	**11.59%**	**311.84**	**88.41%**
MEDIUM	50	TCP	1 Mbps	**960.05**	**25.71%**	**219.38**	**74.29%**
HIGH	150	TCP	2 Mbps	**1923.52**	**34.36%**	**159.99**	**65.64%**

Engineering Analysis:

1. **Contention & Buffer Exhaustion**: As student concurrency scales from 10 to 50 to 150, the packet loss rate climbs sharply from **11.59%** to **25.71%** and peaks at **34.36%**.
2. **PDR Degradation**: Packet Delivery Ratio (PDR) declines steadily from **88.41%** to **74.29%** down to **65.64%**, illustrating severe packet discard at the campus gateway switch buffer.
3. **Throughput Scaling**: Total delivered throughput reaches saturation near the configured bottleneck link capacities (498.27 Kbps out of 512 Kbps; 960.05 Kbps out of 1000 Kbps; 1923.52 Kbps out of 2000 Kbps).

B. Multi-Rate TCP vs UDP NS-2 Results

Bandwidth Rate	TCP Throughput	UDP Throughput	TCP Loss (%)	UDP Loss (%)	TCP PDR (%)	UDP PDR (%)
256 Kbps	249.63 Kbps	251.20 Kbps	46.72%	50.93%	53.28%	49.07%
512 Kbps	499.75 Kbps	449.54 Kbps	31.33%	0.00%	68.67%	100.00%
1 Mbps	960.62 Kbps	857.09 Kbps	26.92%	0.00%	73.08%	100.00%
2 Mbps	1887.37 Kbps	1691.65 Kbps	24.60%	0.00%	75.40%	100.00%
5 Mbps	4559.98 Kbps	4191.23 Kbps	15.28%	0.00%	84.72%	100.00%

C. Advanced Protocol Simulation Findings

- **Go-Back-N ARQ (Agent/TCP Tahoe + ErrorModel)**:
- Total Injected Packets: 301
- Packets Dropped: 18
- Retransmitted Packets: 19
- Demonstrates that when an unacknowledged frame is lost, Tahoe resets `cwnd = 1` and retransmits cumulatively from the missing segment forward.
- **Leaky Bucket Traffic Shaping**:
- **Unshaped Workload**: Bursts instantly saturated the unbuffered link with heavy packet drops.
- **Shaped Workload**: The bounded DropTail queue smoothed the burst and delivered **430.82 Kbps** steadily through the 512 Kbps leak link.

21. Project Limitations

1. **Physical Wireless Channel Modeling**: The project focuses on campus wired distribution trunks (*R_1* → *R_2*) and does not simulate 802.11 Wi-Fi multi-path fading or RF interference.
2. **Dynamic Routing Protocols**: Topologies use static shortest-path routing; OSPF / BGP link-state re-convergence is not modeled.
3. **Hardware Platform Dependence**: Live Mode requires a Linux environment or Windows 10/11 with WSL2 enabled.

22. Future Enhancements

1. **NS-3 Simulation Backend**: Transition from NS-2 to modern C++ NS-3 discrete-event architecture.
2. **Wireless 802.11ac/ax Campus AP Modeling**: Incorporate wireless access point contention and student roaming.
3. **Interactive Topology Builder**: Drag-and-drop canvas for custom multi-server university topologies.
4. **Active Queue Management (CoDel / FQ-CoDel)**: Evaluate modern bufferbloat mitigation algorithms.

23. How to Explain This Project in Viva (Viva Voce Preparation)

Q1: What problem does EduNet Analyzer solve?

> ***Answer:*** *It analyzes how simultaneous student access to university e-learning services (LMS, video streaming, quizzes) causes queue buffer overflow, delay inflation, and packet loss at campus bottleneck links, utilizing authentic discrete-event NS-2 simulation.*

Q2: Why is NS-2 used instead of capturing real Wi-Fi packets?

Answer: *Capturing real Wi-Fi only measures a single laptop's current connection. NS-2 allows us to simulate controlled, reproducible laboratory experiments with 10, 50, 100, or 200 concurrent student nodes, exact queue sizes, and deterministic link delays.*

Q3: What do the nodes in the topology represent?

Answer: *Nodes 0 to N-1 represent student workstations in hostels and labs. Node R_1 represents the Access Router, Node R_2 is the Core Router, and Node N+2 is the central LMS server.*

Q4: How is network congestion generated in the simulation?

Answer: *By setting the aggregate student transmission rate higher than the bottleneck link capacity ($R_1 \leftrightarrow R_2$), causing the router's DropTail queue (25 packets) to fill up and drop packets.*

Q5: How is packet loss measured?

Answer: *By counting lines in the NS-2* `.tr` *trace file starting with event* **d** *(drop), or subtracting total received packets from total sent packets:* $\text{Loss \%} = (\text{Dropped} / \text{Sent}) \times 100$.

Q6: How is throughput calculated?

Answer: *By summing the payload bytes of all packets successfully received at the LMS server (***r** *events at server node), multiplying by 8 to get bits, and dividing by simulation duration.*

Q7: How is End-to-End Delay calculated?

Answer: *By subtracting the departure time of a packet ID at the student node (***+** *event) from its arrival time at the server node (***r** *event), and averaging across all received packets.*

Q8: What is Packet Delivery Ratio (PDR)?

Answer: *PDR is the ratio of packets successfully delivered to the destination compared to the total packets transmitted by sources:* $(\text{Received} / \text{Sent}) \times 100$.

Q9: Why compare TCP and UDP?

Answer: *To demonstrate the fundamental trade-off: TCP provides reliable delivery and backs off during congestion (ideal for exams and file downloads), while UDP streams without retransmission or backoff (ideal for live video, but causes heavy loss during congestion).*

Q10: What is the role of the Sliding Window?

Answer: *It provides flow control by allowing up to N frames to be in-flight without waiting for an individual acknowledgment, maximizing link utilization while preventing receiver buffer overflow.*

Q11: How does Go-Back-N handle packet drops?

Answer: *The receiver discards all subsequent out-of-order packets. When the sender's retransmission timer expires, it rewinds its window and retransmits the lost frame and all subsequent frames.*

Q12: How does Leaky Bucket prevent congestion?

Answer: *It absorbs unpredictable traffic bursts into a buffer tank and discharges packets onto the network at a constant, regulated bit rate.*

Q13: Why is WSL2 used on Windows?

Answer: *NS-2 was built for UNIX/Linux. WSL2 provides a genuine Linux kernel running inside Windows, allowing FastAPI to execute native* **`ns`** *commands with full filesystem path translation.*

Q14: What is the purpose of the `.tr` file?

Answer: *The* **`.tr`** *file is the complete discrete-event trace output generated by NS-2 containing every enqueue, dequeue, receive, and drop event with microsecond timestamps.*

Q15: How did you implement Go-Back-N in NS-2 since there is no Agent/GBN?

Answer: *In NS-2, Go-Back-N is modeled using* **`Agent/TCP`** *(Tahoe) paired with cumulative* **`Agent/TCPSink`***. When a packet drops via an* **`ErrorModel`** *on the bottleneck, the sink rejects subsequent out-of-order frames and sends duplicate ACKs. Upon timeout, Tahoe collapses its congestion window to 1 and retransmits all packets starting from the lost sequence number forward, exactly mirroring Go-Back-N ARQ.*

Q16: How did you implement Leaky Bucket in NS-2, and why is RED not acceptable?

Answer: *Random Early Detection (RED) is an active queue management policy that probabilistically drops packets early to prompt TCP backoff; it does not shape traffic. Leaky Bucket is implemented using an intermediate shaper node with a bounded DropTail queue (capacity = bucket size) connected via a dedicated link drained at a constant leak rate (e.g. 512 Kbps).*

Q17: What do the metrics show as user load scales from 10 to 50 to 150 students?

Answer: *Packet loss increases from 11.59% to 25.71% and up to 34.36%, while PDR drops from 88.41% down to 65.64%. This demonstrates queue buffer exhaustion at the campus access gateway as student demand outstrips bottleneck link bandwidth.*

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